Exercise 11.10

(1) (i) $\frac{1}{4}\log\left|\frac{2+x}{2-x}\right|+c$ (ii) $\frac{1}{20}\log\left|\frac{5+2x}{5-2x}\right|+c$ (iii) $\frac{1}{12}\log\left|\frac{3x-2}{3x+2}\right|+c$

(2) (i) $\frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}-3+x}{\sqrt{2}+3-x}\right|+c$ (ii) $\frac{1}{10}\log\left|\frac{x-4}{x+6}\right|+c$ (iii) $\log\left|x+2+\sqrt{x^2+4x+2}\right|+c$

(3) (i) $\log\left|x+2+\sqrt{(x+2)^2-1}\right|+c$ (ii) $\log\left|x-2+\sqrt{x^2-4x+5}\right|+c$ (iii) $\sin^{-1}\left(\frac{x-4}{5}\right)+c$

Exercise 11.11

(1) (i) $\log\ |x^2+4x-12|-\frac{7}{8}\log\left|\frac{x-2}{x+6}\right|+c$ (ii) $\frac{5}{2}\log|x^2+2x+2|-7\tan^{-1}(x+1)+c$

(iii) $\frac{3}{4}\log|2x^2-2x+3|+\frac{\sqrt{5}}{2}\tan^{-1}\left(\frac{2x-1}{\sqrt{5}}\right)+c$

(2) (i) $5\sin^{-1}\frac{x-2}{\sqrt{13}}-2\sqrt{9+4x-x^2}+c$ (ii) $\sqrt{x^2-1}+2\log\left|x+\sqrt{x^2-1}\right|+c$

(iii) $2\sqrt{x^2+4x+1}-\log\left|x+2+\sqrt{x^2+4x+1}\right|+c$

Exercise 11.12

(1) (i) $\frac{x+1}{2}\sqrt{x^2+2x+10}+\frac{9}{2}\log\left|x+1+\sqrt{x^2+2x+10}\right|+c$

(ii) $\frac{x-1}{2}\sqrt{x^2-2x-3}-2\log\left|x-1+\sqrt{x^2-2x-3}\right|+c$

(iii) $\frac{x-5}{2}\sqrt{10x-x^2-24}+\frac{1}{2}\sin^{-1}(x-5)+c$

(2) (i) $\frac{1}{4}\left[(2x+5)\sqrt{9-(2x+5)^2}+9\sin^{-1}\left(\frac{2x+5}{3}\right)\right]+c$

(ii) $\frac{1}{4}\left[(2x+1)\sqrt{81+(2x+1)^2}+81\log\left|2x+1+\sqrt{81+(2x+1)^2}\right|\right]+c$

(iii) $\frac{x+1}{2}\sqrt{(x+1)^2-4}+2\log\left|x+1+\sqrt{(x+1)^2-4}\right|+c$

Exercise 11.13

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
1	3	4	1	3	1	3	2	4	3	4	2	4	4	1
(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)					
1	3	2	4	1	3	4	3	1	4					

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